

RAM AND STORAGE

1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Standard Speeds	800 – 2133 MT/s	2133 – 3200 MT/s	4800 – 8400+ MT/s
Operating Voltage	1.5 V (Standard) / 1.35 V (DDR3L)	1.2 V	1.1 V
Pin Count (DIMM / SODIMM)	240-pin (Desktop) / 204-pin (Laptop)	288-pin (Desktop) / 260-pin (Laptop)	288-pin (Desktop) / 262-pin (Laptop)
Power Management	Controlled by motherboard	Controlled by motherboard	On-DIMM PMIC (Power Management IC)
Channel Architecture	1 × 64-bit channel per DIMM	1 × 64-bit channel per DIMM	2 × 32-bit independent subchannels per DIMM
Typical Capacity per Stick	2 GB – 8 GB (up to 16 GB rare)	4 GB – 32 GB	16 GB – 64 GB+
Typical Use Cases	Legacy desktop/laptop systems (pre-2015), home media servers, retro consoles.	Mainstream office PCs, mid-range gaming, budget laptops, older workstations.	Modern high-end gaming rigs, 4K/8K video editing, local AI modeling, high-load virtualization.

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2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully

1. **Shut down the computer** and disconnect the power adapter/cable. For a desktop, switch off the power supply.
2. **Open the computer case/back panel** carefully.
3. Locate the **RAM slots** on the motherboard. RAM modules are long, narrow circuit boards with clips at both ends.



4. **Take a clear photo** of the installed RAM module while it is still in the slot.
5. To remove it, gently push the **locking clips outward** on both sides of the RAM slot. The RAM will pop up; hold it by the edges and remove it.
6. Reinstall the RAM by aligning the **notch** on the RAM module with the ridge in the slot. Press down evenly until both clips lock into place.
7. Close the case/back panel and reconnect the power.
8. **Turn on the computer** and verify that it boots normally. You can also check **Task Manager → Performance → Memory** in Windows to confirm that the RAM is detected.

Result: The RAM module was successfully removed and reinstalled. After reconnecting the power, the computer booted successfully and the RAM was detected by the operating system.

3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Storage Type	Key Difference	Speed	Typical Use Case / App
HDD	Uses spinning magnetic disks	Slowest	File storage – storing photos, videos and documents
SSD (SATA)	Uses flash memory and no moving parts	Faster than HDD	Windows/Office – faster boot and application loading
NVMe SSD	Flash storage connected through PCIe	Fastest	Gaming – faster game loading and large-file transfers

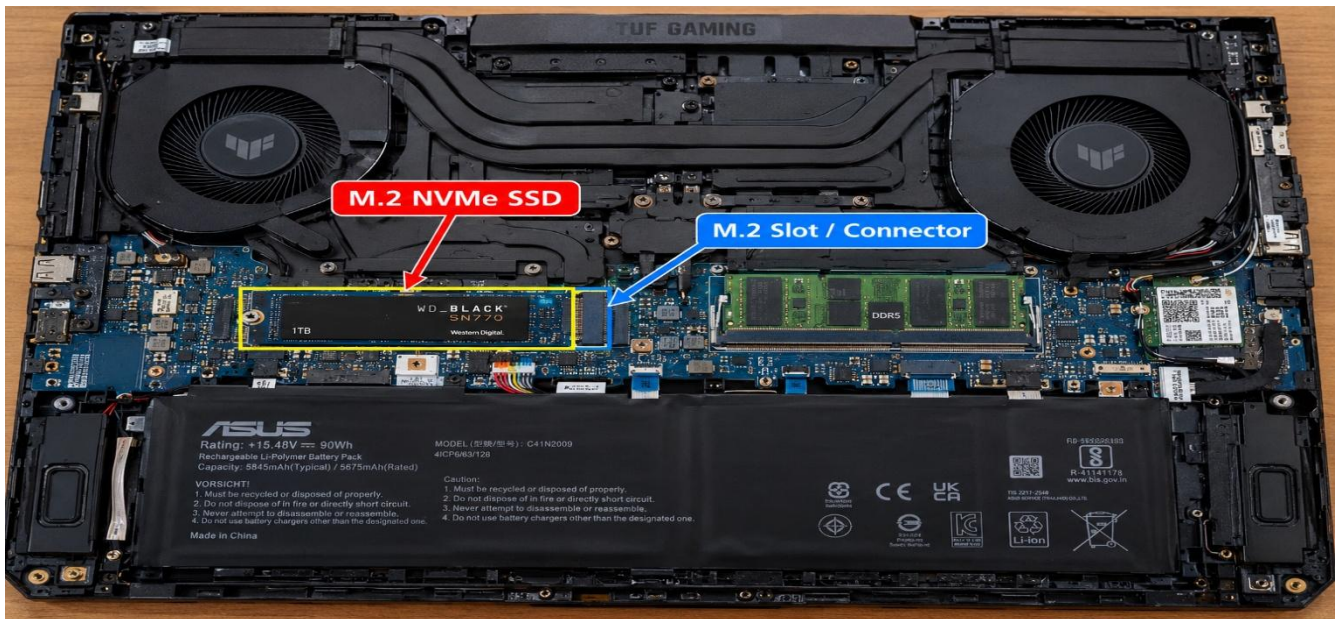
Three main differences

- 1. Technology:** HDD uses spinning disks; SSD and NVMe use flash memory.
- 2. Connection:** SATA SSDs normally use SATA; NVMe drives generally use an **M.2 PCIe/NVMe interface**.
- 3. Performance:** NVMe is generally faster than SATA SSD, while HDD is generally the slowest.

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4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

1. Shut down the computer and disconnect the power.
2. Open the laptop/desktop back panel.
3. Locate the storage drive:
 - **2.5-inch HDD/SSD:** usually connected with a **SATA data + SATA power connector**.
 - **M.2 SSD:** a small circuit board secured directly to the motherboard with a screw.
4. Take a clear photo showing the **drive and its connector**.



5. Carefully disconnect/remove the drive.
6. Reinstall it securely and reconnect all cables.
7. Start the computer and check **File Explorer → This PC** or **Disk Management** to confirm the storage is recognized.

Result: The storage drive was successfully removed and reinstalled. The connector type was identified, and after reassembly the system successfully detected the storage drive.